Worksheet 3

Dispersion and concentration – EXERCISES

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Exercise 1 – Mean absolute deviation, variance, and standard deviation

x_i	2	4	7	12	15
n_i	4	2	5	7	2

1. Calculate the mean.
2. Calculate the mean absolute deviation $e = \frac{1}{N}\sum n_i|x_i - \bar{x}|$.
3. Calculate the variance $V = \frac{1}{N}\sum n_i(x_i - \bar{x})^2$ and standard deviation σ.
4. Calculate the range and coefficient of variation.

Exercise 2 – Dispersion of grouped data

Use the 210-customer willingness-to-pay distribution from Worksheet 2. Complete a table with class midpoints, $n_i c_i$, c_i^2, and $n_i c_i^2$; then calculate variance, standard deviation, coefficient of variation, and interquartile range. **Exercise 3 – Medial value, Lorenz curve, and Gini index**

Monthly salaries are grouped as follows:

Class (hundreds of euros)	$[10, 20[$	$[20, 30[$	$[30, 40[$	$[40, 60[$
Count	20	30	30	20

Calculate class midpoints, masses $n_i c_i$, relative and cumulative masses; find the medial value MI as defined in the course; construct the Lorenz curve; calculate the Gini index using trapezoids.